

hypethesis

Pixie Dust

■ What is it?

Pixie Dust is an indigenous technology that IBM uses to pack data on a portable hard drive. IBM developed this new technology a year ago, and all IBM notebooks integrate this technology into their hard disks.

■ How does it work?

IBM uses a home grown data storage technology called the antiferromagnetically-coupled (AFC) media. Pixie Dust is a three-atom-thick layer of the element Ruthenium. The Ruthenium element forces adjacent layers to orient themselves in magnetically opposite directions. This makes the structure appear much thinner than it actually is—so small that high-density bits can be written easily on it, while retaining the magnetisation due to the media's overall thickness.

■ What can it do?

IBM is now packing five layers of Pixie Dust, allowing up to 70 billion bits of data to be written on each square inch of disk space, up from 40 billion bits. IBM predicts that it will achieve 100 billion bits per square inch in 2003. The Travelstar 80GN series of mobile hard disks have proven reliability with high shock robustness and low power consumption. 8 MB cache makes it the world's highest performing 4200-rpm mobile hard drive.

Internet on the roof

Soon, a hundred lucky households in Cardiff, Wales, will live the broadband dream and, perhaps, a glimpse into the broadband future—movies on demand, broadcast quality video chats and download speeds at 4 Mbps. This new service is based on Mesh radio, a high capacity wireless network. Unlike conventional wireless networking services, this one turns every end user's house into a mini base station. With four directional



antennas that lock onto other antennas, each mini base station achieves nearly a 100 per cent cover. If a station goes down, recovery is easy—come back on and lock into the neighbour's antenna.

Most broadband users in India suffer because there are too many subscribers and too little bandwidth. In this radical new system, Mesh radio will not slow down when a lot of people access it at the same time.

SMS causes divorce

With the rapid acceptance of cell phones by the Indian middle class, the newest cause for concern is marital disharmony, claims the National Human Rights Council. The group burned a cellular phone in protest in a demonstration in New Delhi on November 11, 2002. Subhash Gupta, the group leader, alleges that SMS is against Indian culture and has resulted in a rather large number of divorces.



With all the heat on the ill effects of wireless telephony being focussed on the radiation emitted from mobile devices, this is one of the first protests of its kind against the spread of SMS. The Council argues that the brief, yet effective language used in SMS sparks discord among married couples. Perhaps cause for concern rises from the nature of SMS, where one can chat with just about anybody using a mobile phone.

Playing a new tune

For all MUSICMATCH fans, here is a reason to rejoice—MUSICMATCH Jukebox 7.5 (www.musicmatch.com) has just been released. The latest version brings with it new online jukebox technology that allows for effortless transfer to portable devices, lesser time to create custom CDs and advanced music tagging and labelling that will help you quickly and effortlessly add track information for digital audio files using the company's Radio MX subscription service. It also comes with a sleeker, easy-to-use interface.

Though MUSICMATCH decided to stop supporting the Mac version late in October, this one carries support for quick-syncing the Apple iPod for Windows operating systems. It still has syncing support for all the popular portable MP3 players, including the Nomad Jukebox. Version 7.5 also lets you create full-colour CD labels and jewel case inserts for all of your custom CDs.

snapshot

89% of international e-commerce users base their buying decisions on transaction security, while **59%** base theirs on price

Source: WorldPay eCommerce Survey

■ New research proves computing causes strain and lethargy ■ Mitch Kapor, co-founder of Lotus Development, to build a free, open-source equivalent of MS Outlook